

USER MANUAL

Cell-Free Protein
Synthesis Kit

Standard 50 µL reaction · T7 · Batch

C1110 · 100 rxn

C1120 · 200 rxn

For research use only. Not for use in diagnostic or therapeutic procedures.

00 Product overview

OVERVIEW

The ZerothBIO Cell-Free Protein Synthesis Kit enables coupled in vitro transcription and translation directly from user-supplied template DNA in an E. coli-based cell-free system. Mix **2x Master Mix** and **E. coli Extract** with a plasmid or PCR product compatible with the T7 expression system, and adjust the final volume with nuclease-free water.

T7 RNA polymerase transcribes the template into mRNA, which is then translated into the target protein by the ribosomes, tRNAs, amino acids, and accessory factors in the E. coli Extract. This manual covers the two current commercial pack sizes listed below, for the 50 µL standard reaction only.

00.1 At a glance

EXPRESSION SYSTEM T7 · Batch Coupled transcription–translation	REACTION 50 µL Total volume per reaction	INCUBATION OPTIONS 30 °C for 3 h or 37 °C for 2 h
---	---	--

00.2 Kit contents

Component	C1110 · 100 rxn	C1120 · 200 rxn
2x Master Mix	5 vials × 500 µL	5 vials × 1,000 µL
E. coli Extract	5 vials × 300 µL	5 vials × 600 µL
Reaction capacity	50 µL × 100 rxn	50 µL × 200 rxn

Template DNA, nuclease-free water, reaction tubes and analysis materials are supplied by the user and are not kit components. Reaction capacity describes pack size; it does not specify a plate format.

00.3 How this manual is organized

01	Before you begin	Storage, materials, template DNA, thawing
02	Set up the reaction	Composition, mixing, incubation, run record
03	Analyze and document expression	Analysis method, specifications, troubleshooting
04	Ordering and technical support	Cat. No., contact, document control

READ BEFORE STARTING

Before you begin (storage, materials and template DNA), **Set up the reaction**, and **Analyze and document expression**.

01 Before you begin

PREPARATION

01.1 Storage and handling

STORAGE

–70 °C to –20 °C

All components

Store all components at –70 °C to –20 °C. **Minimize freeze-thaw cycles for the E. coli Extract**, while the 2x Master Mix is more resilient to repeated thawing. Thaw all components on ice immediately before use.

Upon receipt, aliquoting the E. coli Extract is **strongly** recommended to reduce repeated freeze-thaw exposure; store the aliquots at –70 °C. Use the product by the EXP date indicated on the product label, and record the Lot No. for each run.

SAFETY

Consult the current MSDS for 2x Master Mix and E. coli Extract before handling. This manual provides experimental instructions and does not replace the MSDS.

01.2 Prepare your materials

User-supplied item	Purpose
Template DNA	Plasmid or linear PCR product encoding the target protein
Nuclease-free water	Brings Template DNA + water to 10 µL per 50 µL reaction
Calibrated pipettes and nuclease-free filter tips	Dispense the reaction components
Reaction tubes and ice or cooling block	Prepare and mix reactions on ice
Incubator, heating block or thermal cycler	Hold 30 °C or 37 °C for the selected incubation option
Microcentrifuge and analysis reagents	Brief spin-down; SDS-PAGE, Western blot or bioactivity assay

01.3 Check the DNA template

The template must contain a **T7 promoter**, **ribosome binding site (RBS)** and **T7 terminator**. Circular plasmid DNA and linear PCR products may both be used.

Recommended starting concentration: approximately 5–8 µg/mL final in the reaction mixture, i.e. 0.25–0.4 µg of template DNA per 50 µL reaction. Choose the template volume (x µL) from your stock concentration and optimize for each target.

EXAMPLE

A 100 µg/mL stock gives 0.25–0.4 µg in 2.5–4 µL. Because Template DNA + water is limited to 10 µL, a stock below 25 µg/mL cannot reach 0.25 µg per reaction.

01.4 Thaw and resuspend

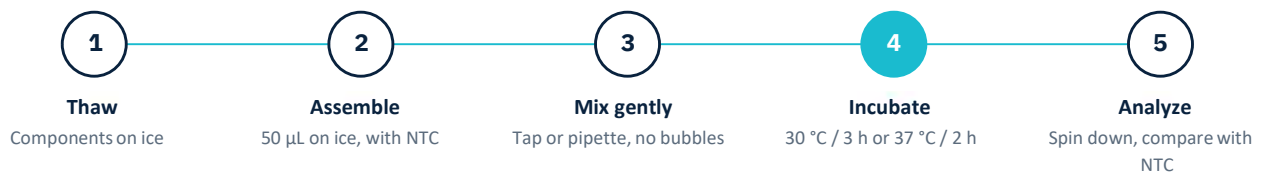
- 1 Thaw 2x Master Mix and E. coli Extract on ice.
- 2 Briefly spin down all tubes and return them to ice.
- 3 Ensure both components are fully and uniformly resuspended. Avoid generating bubbles in E. coli Extract.

KEEP ON ICE

Keep components on ice during reaction preparation.

02 Set up the reaction

PROTOCOL



Prepare each reaction on ice. Include a **no-template control (NTC)** prepared with the same components but without template DNA.

02.1 Standard reaction composition

2x Master Mix · 25 µL	E. coli Extract · 15 µL	DNA + water · 10 µL
0	25	40
		50 µL

Component	No-template control	Expression reaction
2x Master Mix	25 µL	25 µL
E. coli Extract	15 µL	15 µL
Template DNA	—	x µL
Nuclease-free water	10 µL	(10 – x) µL
Total volume	50 µL	50 µL

x is the volume of template DNA solution (see 01.3 Check the DNA template). Template DNA and nuclease-free water must total 10 µL so that the total reaction volume remains 50 µL; for an expression reaction, $0 < x \leq 10 \mu\text{L}$.

02.2 Mix gently

Gently mix by tapping the tube or pipetting. Keep the reaction on ice until ready for incubation and avoid introducing bubbles into the extract.

FINAL CHECK

Confirm the **50 µL total volume**, label the **no-template control** and record the **selected incubation condition** before starting.

02.3 Incubate for protein expression

Option	Temperature	Duration
A	30 °C	3 h
B	37 °C	2 h

Alternatives, not consecutive steps. Use either option.

After incubation, briefly centrifuge the completed reaction mixture before analysis.

02.4 Record the run

Record the kit Cat. No. and Lot No., template identity, DNA stock concentration, DNA amount per reaction, DNA and water volumes, incubation temperature and duration, and the selected assay. Retain the no-template control result with the expression result.

- | | | |
|---|--|---|
| ☐ Kit Cat. No. and Lot No. | ☐ DNA amount per reaction | ☐ Selected assay |
| ☐ Template identity | ☐ DNA and water volumes | ☐ NTC result retained with expression result |
| ☐ DNA stock concentration | ☐ Incubation temperature and duration | |

03 Analyze and document expression

ANALYSIS

03.1 Select the analysis method

Method	Instruction
SDS-PAGE	Load 2 µL of the CFPS reaction mixture as a starting point; include an NTC.
Western blot	Load 2 µL of the CFPS reaction mixture as a starting point; include an NTC.
Relevant bioactivity assay	Use an assay appropriate to the target protein; determine target concentration with a suitable method.

COMPARE WITH THE NTC

Compare expression samples with the no-template control. The extract contributes background proteins, so a band or bulk protein signal alone should not be treated as target-specific quantification. Use the chosen assay to establish target identity, abundance or activity, as appropriate.

03.2 Interpret the product specifications

Specification	Approved value	Interpretation
Target protein size	≤150 kDa	Listed scope; not a guarantee of successful expression for every target.
Protein yield	0.2–2.0 mg/mL (protein-dependent)	Target-dependent; not an expected value for every construct. Determine expression empirically for each target.
Purification	Not included	Arrange a separate purification workflow if required.

YIELD IS TARGET-DEPENDENT

For bioactivity assays, determine the target concentration using an analysis method appropriate for the target. Do not assume that the full 0.2–2.0 mg/mL range applies to every construct.

03.3 Troubleshooting

Observation	Possible cause	Recommended action
No or low yield	Template lacks T7 promoter, RBS or T7 terminator	Verify the template design and sequence.
	Template DNA outside the recommended starting range	Start at approximately 5–8 µg/mL final (0.25–0.4 µg per 50 µL reaction) and optimize for the target.
	Component activity reduced by repeated freeze-thaw or improper thawing	Use a fresh aliquot; thaw on ice and avoid repeated freeze-thaw cycles.
Target band difficult to distinguish	Background proteins from E. coli Extract	Run an NTC in parallel and confirm with Western blot or another suitable analysis.
Inconsistent reaction volume	Template DNA and water not balanced	Keep Template DNA + water at 10 µL and the total reaction volume at 50 µL.

IF THE ISSUE PERSISTS

Contact ZerothBIO technical support with the run record (Cat. No., Lot No., template, DNA amount, incubation condition) and the NTC result before changing the protocol.

04 Ordering and technical support

SUPPORT

Cat. No.	Product
C1110	ZerothBIO Cell-Free Protein Synthesis Kit 100 rxn
C1120	ZerothBIO Cell-Free Protein Synthesis Kit 200 rxn

COMPANY

ZerothBIO Inc. | (주)제로스바이오

Suite 1208, Building A, 13, Sinwon-ro 250beon-gil,
Yeongtong-gu, Suwon-si, Gyeonggi-do, Republic of Korea

CONTACT

Email zerothbio@gmail.com

Tel +82-10-9118-1061

Website www.zerothbio.com

DC Document control

Field	Value
Document ID	ZBIO-PRD-CFPS-MANUAL-001
Version / status	v1.3 Renewal · Review Draft R2 — not approved for distribution
Supersedes (on approval)	v1.1.2 Controlled Revision (2026-08-10)
Controlled references	ZBIO-PRD-MASTER-001 v1.4 · ZBIO-PRD-CFPS-TDS-001 v1.0 · ZBIO-PRD-CFPS-MSDS-001 v1.0 · ZBIO-PRD-CFPS-MSDS-002 v1.0
Scope	C1110 (100 rxn) and C1120 (200 rxn), 50 µL standard commercial reaction only.
Technical change vs v1.1.2	None. Layout, wording and user guidance only; all technical values unchanged.

ZEROOTH BIO

Before the First, There's Zeroth



ORDERING

C1110

ZerothBIO Cell-Free Protein Synthesis Kit
100 rxn

C1120

ZerothBIO Cell-Free Protein Synthesis Kit
200 rxn

TECHNICAL SUPPORT

ZerothBIO Inc. | (주)제로스바이오

Email zerothbio@gmail.com · Tel +82-70-8064-1061 · www.zerothbio.com

Suite 1208, Building A, 13, Sinwon-ro 250beon-gil, Yeongtong-gu, Suwon-si, Gyeonggi-do, Republic of Korea